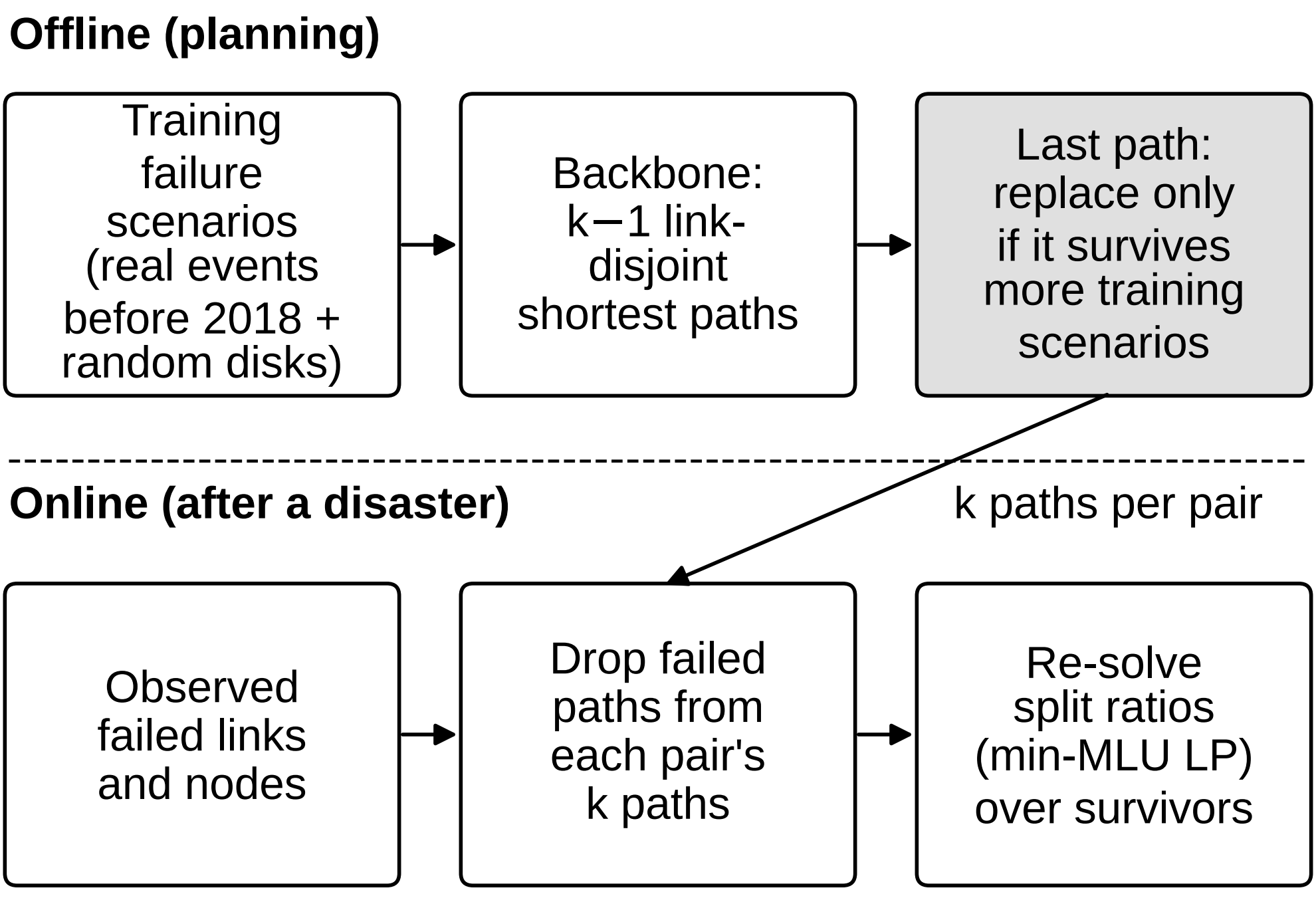
Offline (planning)
Training failure scenarios (real events before 2018 + random disks)
Backbone: k−1 link-disjoint shortest paths
Last path: replace only if it survives more training scenarios
Online (after a disaster)
k paths per pair
Observed failed links and nodes
Drop failed paths from each pair's k paths
Re-solve split ratios (min-MLU LP) over survivors